



SOFTWARE REQUIREMENTS DOCUMENT

BANK OF JARVIS TRADING APPLICATION

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1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for the **Bank of Jarvis Trading Application**— a full-stack web and mobile application that enables clients to independently buy and sell stocks without requiring an in-person meeting with a broker.

1.2 Background

Currently, **Bank of Jarvis** clients must schedule a meeting with a broker to execute stock trades. This manual process creates friction, reduces scalability, and limits client autonomy.

This project aims to:

- Digitize stock trading for clients
- Increase trading accessibility
- Improve operational efficiency
- Migrate 20% of existing brokerage clients to self-directed trading within 12 months of launch

1.3 Go-Live Target

- **Time to production:** 12 months from project start
- **Target SLA:** 99.4% uptime

2. Scope

The **Bank of Jarvis Trading Application** will enable customers to engage in self-directed equity trading through a dedicated mobile platform. The solution will provide secure user authentication, real-time access to market data, order placement and execution capabilities, portfolio tracking, transaction history visibility, and funding integration with existing Bank of Jarvis accounts.

The trading platform will operate as a standalone product rather than as an extension of Bank of Jarvis's primary consumer banking application. However, it will leverage core backend services within the bank's existing technology ecosystem — including identity management, authentication, and compliance frameworks — to maintain alignment with each customer's established financial profile and regulatory obligations.

2.1 In Scope

The complete application is intended to support the following capabilities:

- Secure sign-in using existing Bank of Jarvis online banking credentials, enhanced with multi-factor authentication for added protection.
- Access to comprehensive real-time market data display (price, ticker, volume)
- Portfolio overview dashboard
- Complete transaction records with detailed trade confirmations and searchable historical activity.
- Real-time notifications for order status updates, trade confirmations, and other key account events.
- Integration with core banking and brokerage systems

2.2 Out of Scope

The following items are **not** included in the initial release of the product but may be considered for future phases:

- Options trading
- Crypto trading
- Margin trading
- Desktop or web versions of the platform
- Advanced charting tools

3. Stakeholders

The key stakeholders for the Bank of Jarvis Trading Application include:

- **Bank of Jarvis Customers:** Primary users seeking to trade without setting up broker meetings.
- **Brokers:** Shift focus from routine order entry to high-level advisory services.
- **Product Owner / Business Sponsor:** Responsible for defining business objectives, approving features, and ensuring alignment with Bank of Jarvis' strategic goals
- **Internal IT/Engineering:** Responsible for integrating the internal stock API and banking core.
- **Compliance & Legal:** Ensuring self-directed trades meet regulatory standards.

4. Functional Requirements

4.1 User Authentication and Security

- Users shall authenticate using their existing Bank of Jarvis account credentials.
- Mandatory biometric verification (FaceID/TouchID) for secure access.
- All financial and personal data must be encrypted in transit and at rest.e access.
- Automatic session timeout after 15 minutes of inactivity.

Performance/Latency :

- Login authentication response time should be **under 2 seconds** under normal load.
- Biometric verification should complete within **1 second**.

Error Handling :

If authentication fails:

- Users will receive a clear error message indicating incorrect credentials.
- After **5 failed login attempts**, the account will be temporarily locked for security.
- If biometric authentication fails, users will be prompted to authenticate using their password and MFA.
- If the authentication service becomes unavailable, users will receive a “**Service temporarily unavailable**” message and will be prompted to retry later.

4.2 Dashboard (Post-Login)

4.2.1 Portfolio Overview Tab

- Total portfolio value
- Daily gain/loss
- Total return
- Holdings summary

Each holding must display

- Stock ticker
- Company name
- Quantity owned
- Current price
- Market value
- Day change %

Performance / Latency :

- Portfolio values must refresh within **3 seconds** of a completed trade.
- Market price updates must occur within **5 seconds** of market data updates.

Error Handling :

- If market data is unavailable, the last known price will be displayed with a timestamp indicator.
- If portfolio services fail, users will see a message indicating portfolio information is **temporarily unavailable**

4.2.2 Trade History Tab

- Historical trades
- Filter by date range
- Filter by stock
- View order status (filled, pending, canceled)

Performance / Latency :

- Trade history queries should return results within **2 seconds**.

Error Handling :

- If transaction records cannot be retrieved, users will see a **temporary service interruption notice**.
- If filtering fails, the system will default to displaying the most recent trades.

4.2.3 Search & Discover Tab

- Search bar to search by: Company name, Ticker symbol
- View stock details: Ticker, Current price, Daily change, Trading volume, Market cap
- Basic price chart (1D, 1W, 1M, 1Y)

Performance / Latency :

- Search results must appear within **1.5 seconds**.
- Market data updates must **not exceed 2–3 seconds** latency during market hours.

Error Handling :

- If no results are found, the system must display **“No matching stocks found.”**
- If the market data API fails, the application will notify users that market data is temporarily unavailable.
- Partial results may still be displayed if cached data exists.

4.3 Trading (Buy/Sell)

4.3.1 Order Entry

When placing a trade, the system must display:

- Stock ticker
- Current price (real-time or near real-time)
- Trading volume
- Estimated total cost

Performance / Latency :

- Order entry screen must load within **2 seconds**.
- Market price information must refresh within **2–5 seconds**.

Error Handling :

- If market price data is unavailable, order placement will be temporarily disabled.
- Users will be notified if the system cannot retrieve real-time pricing.

4.3.2 Validation Rules

- Check sufficient funds
- Check trading restrictions
- Compliance validations
- Real-time risk checks

Performance / Latency :

- Order validation should complete within **1 second**.

Error Handling :

If validation fails:

- Users will receive a specific error message indicating the issue, such as:, **Insufficient funds, Trading restrictions, Compliance restrictions**.
- Orders will not be submitted until all conditions are satisfied.

4.3.3 Order Confirmation

- Show estimated execution price
- Show transaction fees
- Require confirmation

Performance / Latency :

- Confirmation processing must complete within **1 second**.

Error Handling :

- If the confirmation service fails, the trade will not be submitted.
- Users will receive an error message prompting them to retry.

4.3.4 Post Trade

- Display confirmation
- Update portfolio instantly (pending state)
- Send email confirmation

Performance / Latency :

- Trade confirmation must appear within **2 seconds** of order submission.
- Portfolio updates should reflect the trade within **3 seconds**.

Error Handling :

- If confirmation delivery fails, the trade record will still be saved and visible in Trade History.
- If email delivery fails, users may still access confirmation through the platform.

4.4 User Settings

Users will have the ability to modify their personal information, manage connected financial accounts, activate or disable biometric authentication, and customize their notification settings and communication preferences.

Performance / Latency :

- Settings updates should process within **1–2 seconds**.

Error Handling :

- If profile updates fail, the system will notify the user and prevent partial updates.
- If notification preferences cannot be saved, users will receive a **retry option**.

5. Non-Functional Requirements

5.1 Performance

- The application must display key interfaces (dashboard, portfolio view, and trading screen) within **2 seconds** during standard operating conditions.
- Order placement requests should be processed and return a response within **1 second** under typical network performance.
- Market data updates must refresh at a minimum interval of **every 5 seconds**

5.2 Availability and Reliability

- The system shall maintain an availability target of 99.4% uptime, excluding planned maintenance windows.
- Trading services must operate reliably and without performance degradation during peak market hours (9.30 AM to 4 PM EST)

5.3 Scalability

- The platform must accommodate growing user demand while maintaining consistent performance levels.
- The system architecture should be designed to scale horizontally to handle increased volumes of market data processing, trade execution traffic, and authentication requests.

5.4 Security

- Ensure that all information is protected through encryption during transmission and while stored (AES-256).
- Enforce login attempt throttling to guard against credential-stuffing attacks.
- Maintain audit and compliance logs for every trade-related operation.
- Implement biometric authentication in accordance with platform-specific guidelines

5.5 Compliance & Regulatory

- The system shall comply with all applicable AML, and trading regulations in Canada
- Every trade must be recorded with a timestamp, user identifier, and full order details to support auditing requirements.
- Management and processing of customer data must adhere to relevant privacy laws and regulations

5.6 Constraints & Timeline

- The application must be fully operational and "Live" within 1 year.
- Mobile-first application (iOS and Android).
- Must consume data from the internal Bank of Jarvis stock API.

5.7 Deployment Plan

The project will be executed in four phases over a 12-month period.

- Phase 1 (Months 1–3) will focus on finalizing the system architecture, selecting vendors, and completing a thorough compliance review.
- Phase 2 (Months 4–8) will cover core application development, integration with the Internal Stock API.
- Phase 3 (Months 9–10) will involve user acceptance testing (UAT), security validation, and load testing to ensure the platform meets performance and reliability requirements.
- Finally, Phase 4 (Months 11–12) will focus on delivering a proof of concept to a selected group of clients, demonstrating the application's core capabilities and gathering feedback to inform future enhancements.

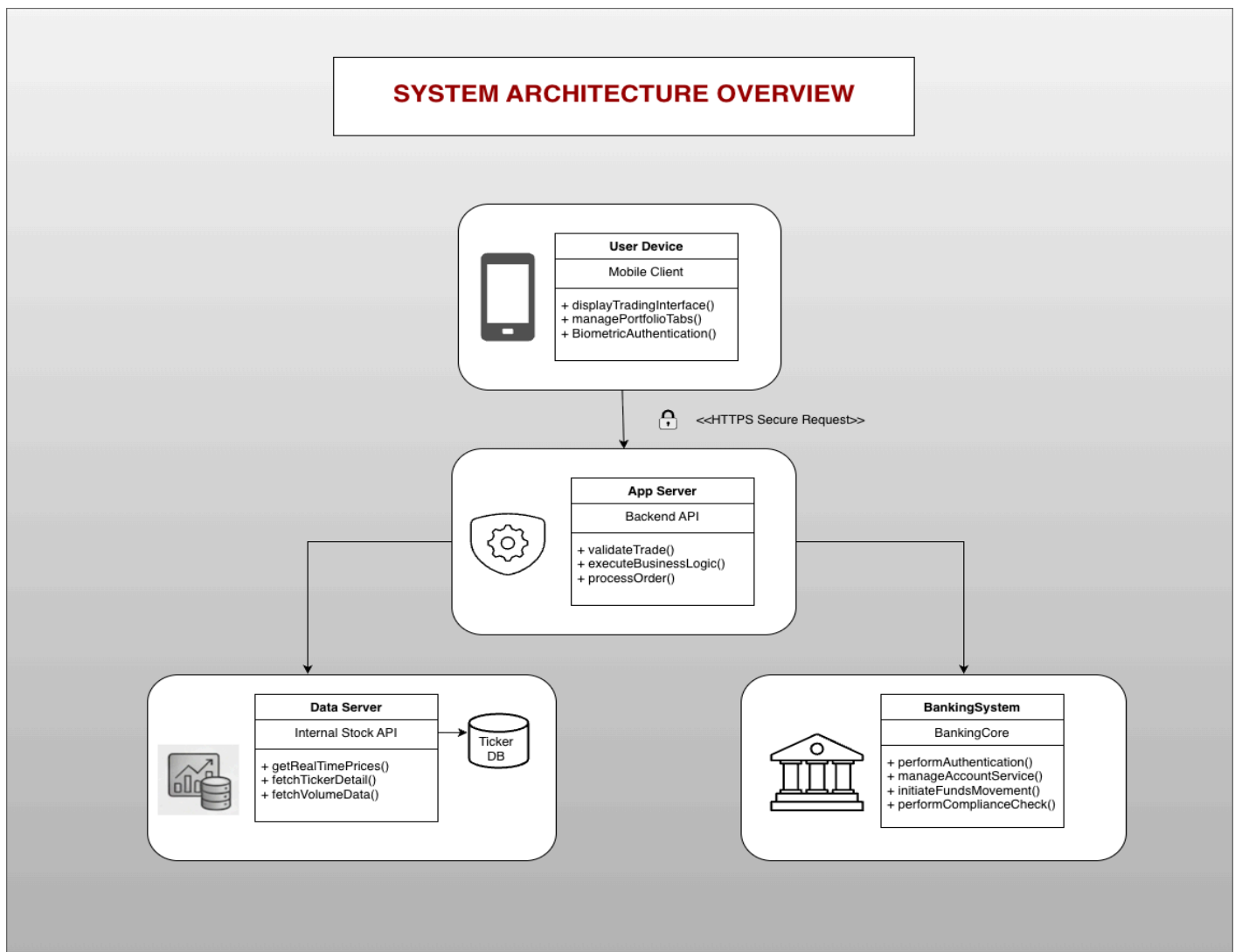
5.8 Business Target

The platform is designed to migrate **at least 20%** of the current client base from manual broker-assisted trading to digital self-service.

6. System Architecture Overview

The application follows a client-server model where the mobile frontend communicates with a secure backend orchestrator.

- Mobile Client : UI for trading, portfolio tabs, and biometric capture.
- Backend API : Validates trades and interfaces with the internal stock API.
- Internal Stock API : Source for real-time price, ticker, and volume data.
- Banking Core : Handles authentication and movement of funds.



7. Use Cases

This section outlines the key actions users take within the Bank of Jarvis Trading Application. Each use case illustrates a complete scenario, detailing how users accomplish particular objectives. Scenarios cover activities such as signing in, checking portfolios, executing stock trades, and tracking account activity. The steps emphasize standard user workflows, highlighting security, compliance, and ease of use.

Use Case ID	Use Case Name	Description	Flow of Events
UC-01	User Login	Access via bank credentials with biometric MFA.	1. User opens app. 2. User enters Bank of Jarvis credentials. 3. System prompts for biometric scan. 4. User provides FaceID/TouchID. 5. System validates credentials and MFA via Banking Core. 6. User is redirected to Portfolio dashboard
UC-02	Market Data Search	View ticker, price, and volume via internal API.	1. User navigates to the "Search" tab. 2. User enters a ticker or company name. 3. System queries Internal Stock API. 4. System displays real-time price, symbol, and trading volume.
UC-03	Place Buy/Sell Order	Execute whole-share trades with automated funding.	1. User selects "Buy" or "Sell" on a specific ticker. 2. System validates Eligibility. 3. User enters quantity 4. The system calculates total cost based on Internal API price. 5. Backend checks for sufficient funds. 6. User confirms; System executes trade and settles funds.
UC-04	View Portfolio	Monitor holdings, allocation charts, and performance.	1. User selects the "Portfolio" tab.

			2. System retrieves current holdings and cash balance from the database. 3. System aggregates real-time valuations via Internal Stock API. 4. System renders allocation charts and performance metrics.
UC-05	Check History	Review chronological logs of all past trade activity.	1. User selects "History" tab. 2. System queries the transaction database for the user's ID. 3. System displays a list of past orders including date, ticker, type (Buy/Sell), and total amount.

8. Conclusion / Summary

In summary, the Bank of Jarvis Trading Application will deliver a secure, scalable, and user-friendly platform for self-directed equity trading. It integrates portfolio management, real-time market data, order execution, and compliance services, providing customers with a seamless trading experience while maintaining regulatory and operational standards.

9. Next Steps

With these requirements defined, the next phase involves detailed system design, development, and testing to ensure compliance, performance, and usability objectives are met. A phased rollout strategy will enable a smooth launch, with continuous monitoring to achieve the target SLA and client adoption goals.